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Article · January 1994

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The Research Base for APA's Learner-Centered Psychological Principles

Patricia A. Alexander and P. Karen Murphy

Between 1990 and 1996, an impressive collection of educational researchers contributed their wisdom and years of well-honed insights to a set of 14 psychological principles that they hoped would guide the redesign and reform of American schools (e.g., American Psychological Association [APA] Presidential Task Force on Psychology in Education and Mid-Continent Regional Educational Laboratory [McREL], 1993). These psychological principles and the philosophical statement framing them have subsequently been revised and are currently under review (APA Board of Educational Affairs, 1995). In our view, these resulting principles can be invaluable tools for practicing professionals as they seek to understand and facilitate the learning of others.

Our role in this critical enterprise began several years ago. At that time, we were asked to offer an analysis of the literatures in learning and instruction, motivation, and development that might underlie the initial set of 12 principles forwarded in the original APA–McREL document (Alexander & Murphy, 1994). Thankfully, during the past decade, Alexander and colleagues (Alexander & Judy, 1988; Alexander & Kulikowich, 1994; Alexander, Kulikowich, & Jetton, 1994; Alexander, Parsons, & Nash, 1996; Alexander, Schallert, & Hare, 1991) engaged in multidimensional reviews of the literature that have synthesized research in these same areas (i.e., learning and instruction, motivation, and development). As we illustrate in this chapter, the general premises

A version of this chapter was presented at the annual meeting of the American Educational Research Association, New Orleans, April 1994.

or postulates that were formulated in these reviews provide strong support for many of the ideas represented in both the original and revised learner-centered principles, described by Lambert and McCombs in the opening chapter of this volume (see chap. 1).

To perform our review of these literatures, we took an initial step of distilling the original 12 statements offered by the task force into 5 more general statements (see Exhibit 1), which we felt could meaningfully serve educators at all instructional levels. We then proceeded to identify bodies of research that we felt were relevant to these general premises. Further, we specified emerging psychological issues that we found missing or only weakly addressed in the original 12 principles outlined by the task force. The resulting research review, which we present in this chapter, served as one of the sources of information that APA's Board of Educational Affairs used in its preparation of the revised learner-centered psychological principles (APA, 1995). For this reason, several of the missing or weakly addressed areas of concern we describe in this chapter are now evident or more strongly articulated in the revised 14 principles. For the reader's convenience, our discussion of the 5 guiding premises makes reference to the pertinent principles as stated

Exhibit 1

Five General Statements Related to the Learner-Centered Principles

The knowledge base

One's existing knowledge serves as the foundation of all future learning by guiding organization and representations, by serving as a basis of association with new information, and by coloring and filtering all new experiences.

Strategic processing or executive control

The ability to reflect on and regulate one's thoughts and behaviors is essential to learning and development.

Motivation and affect

Motivational or affective factors, such as intrinsic motivation, attributions for learning, and personal goals, along with the motivational characteristics of learning tasks, play a significant role in the learning process.

Development and individual differences

Learning, although ultimately a unique adventure for all, progresses through various common stages of development influenced by both inherited and experiential/environmental factors.

Situation or context

Learning is as much a socially shared undertaking as it is an individually constructed enterprise.

in the latest revision. In addition, we have noted any significant deviation between our initial concern (e.g., lack of a specific principle related to assessment) and the modifications that are reflected in the document prepared by the Board of Educational Affairs.

We conclude our literature survey in this chapter by considering the potential impact of these psychological premises on future educational practices—the explicit goal of the APA task force. Before we proceed, however, it is important to place the subsequent analysis within an appropriate theoretical and philosophical context. First, we heartily concur with the mission of the task force to infuse knowledge of human learning, motivation, and development into ongoing efforts to improve the educational experience of all people. Educational reform has too long overlooked or undervalued the beneficial knowledge that can be contributed by research in psychology and related fields. In the judgment of Alexander and Knight (1993), this devaluing of psychological processes has been a contributing factor to the failure of reform movements in education to bring about meaningful progress. Instead, basic issues are updated and recycled, often without regard to their philosophical and psychological roots (Alexander, Murphy, & Woods, 1996).

Second, as with any collaborative effort, even one as effective as the work of the aforementioned task force and the Board of Educational Affairs, the resulting set of principles must be recognized as a compromise among members who came to the task with their own research interests and constructions. This is not to question, in any way, the outcome of these renowned researchers' efforts, but only to recognize that no committee report results without negotiation, collaboration, and ultimately compromise. This observation is one reason why we have chosen to focus our analysis on the underlying tenets espoused in the principles rather than concentrate on their specific language or phrasing.

Third, the guidelines offered by the task force are, indeed, principles that are still undergoing review and revision. As with any "living" document grounded in research that is continually developing, such ongoing review and revision are to be welcomed and encouraged. Consequently, these espoused principles should not be viewed as prescriptions that ensure or even promise effective educational redesign or reformation. Nothing short of long-term, concerted, and cooperative effort on the part of all partners in the educational enterprise can achieve such a laudable goal.

A Reframing of the Learner-Centered Principles

To frame the contents of the original 12 psychological principles proposed by the task force, within the existing literature, we found it helpful to distill them into 5 more general statements that pertain to broad dimensions of learning. By formulating fewer and more general statements, we were better able to specify a body of recent works that substantiates the gist of the original principles. In essence, our goal, as mentioned earlier, was to evaluate the general intention of the learner-centered principles rather than focus our energies on assessing their exact wording or phrasing.

Parsed down to their fundamental roots, we hold that the learner-centered principles relate to five essential dimensions of meaningful learning that have been systematically investigated in psychology and related disciplines for decades. Those dimensions are

1. the knowledge base,
2. strategic processing or executive control,
3. motivation and affect,
4. development and individual differences, and
5. situation or context.

Specifically, it is our judgment that the original 12 statements, as well as the revised 14 principles, can be combined and simplified into these 5 premises, presented in Exhibit 1.

The Knowledge Base

One's existing knowledge serves as the foundation of all future learning by guiding organization and representations, by serving as a basis of association with new information, and by coloring and filtering all new experiences.

Several of the principles formulated by the task force (e.g., Principles 1, 2, and 3) speak to one of the most powerful and consistent findings to emerge from the research in cognition over the past several decades. Simply stated, the knowledge that learners possess becomes an extremely powerful force, not only in what information they attend to (e.g., Anderson, Pichert, & Shirey, 1983; Hidi, 1990; Reynolds & Shirey, 1988) or how that information is perceived (e.g., Gibson, 1966) but also in what is judged by learners as relevant or important (e.g., Alexander, Jetton, Kulikowich, & Woehler, 1994; Pichert & Anderson, 1977; Schraw

& Dennison, 1994) and what is comprehended and remembered (e.g., Alvermann, Smith, & Readence, 1985; Anderson, Reynolds, Schallert, & Goetz, 1977; Pritchard, 1990). In essence, the knowledge base is the total of all an individual knows or believes (Alexander et al., 1991). Truly one's knowledge base is the scaffold that supports the construction of all future learning (Principle 3). Further, because each individual's knowledge base is unique, so too are his or her constructions within that scaffold (Principle 12).

Through the 1970s and into the 1980s, a great flood of research appeared concomitantly with the rise of cognitive psychology that demonstrated how prior knowledge was predictive of learning outcomes (e.g., Alexander, 1996; Anderson, 1977; Bransford & Franks, 1972). Classic studies during this period further demonstrated how the organization and structure of that knowledge often differentiated novice learners from those with greater expertise (e.g., Chi, Feltovich, & Glaser, 1981; Larkin, McDermott, Simon, & Simon, 1980). However, as research on the knowledge base matured, several changes in understanding about the nature and role of that knowledge occurred.

First, since the early information-processing studies, knowledge has come to be viewed as a multifaceted construct that encompasses many interactive dimensions (Alexander et al., 1991), including sociocultural knowledge (Principles 11 and 13), strategic abilities (Principle 4), personal beliefs (Principle 13), and goals (Principle 2). For example, Lipson (1983) examined how students' religious orientation influenced the way they comprehended and recalled passages about religious practices, whereas Chambliss and Garner (1996) investigated how adults' views on logging practices in the western United States altered their reading of persuasive texts. In addition, Alexander and Kulikowich (1991) found that readers' knowledge about human biology was a significant factor in their strategic processing of analogies embedded in exposition. Thus, although researchers have continued to study the knowledge base and substantiate its powerful role in subsequent learning, they conceive of it in an increasingly more complex, multidimensional way (Alexander, Murphy, Woods, Duhon, & Parker, 1997).

Sometimes the dimensions of one's knowledge base, just described, work in concert. At other times, however, these dimensions operate in conflict with one another, as when one's informal knowledge or spontaneous concepts are in opposition to the formal or scientific concepts that are the mainstay of the instructional enterprise (e.g., Alexander, 1992; Gardner, 1991; Vygotsky, 1934/1986). Further, the knowledge

base is not constructed solely of factual or schooled content stored and organized in memory, knowledge that can be retrieved and reflected on (Prawat, 1989), or knowledge that can be externally validated (Alexander & Dochy, 1994, 1995; Mayer, this volume, chap. 14). Instead, it is accepted that the knowledge learners possess is often acquired from nonacademic life experiences (Gardner, 1991) and is, thus, a unique reflection of their sociocultural heritage (Principle 13).

Also, this knowledge can remain buried in memory, as in the case of implicit theories (e.g., Alexander, 1992; Gardner, 1991; Prawat, 1989; Sternberg, Conway, Ketron, & Bernstein, 1981), or can be seriously flawed or misleading (e.g., Alvermann et al., 1985; Champagne, Klover, & Anderson, 1980; Marshall, in press). Therefore, although knowledge is generally held to be a positive force in the learning process, it is no longer judged to always be so (Principle 3). Because learners operate on the incomplete and inaccurate knowledge they possess, their existing knowledge and the misconceptions that are part of it can actually deter or interfere with future growth (e.g., Chinn & Brewer, 1993; Perkins & Simmons, 1988). Consequently, educators must be aware of the misconceptions that students may commonly hold in an area of study and must work to challenge those misconceptions within the context of meaningful instruction (Roth, 1985).

Finally, another change that has occurred in the study of the knowledge base over the past 30 years is the recognition that knowledge, particularly as it relates to academic learning, is associated with specific fields of studies (i.e., domains) and is applied differentially on the basis of the characteristics of the learning task or the context in which it is situated (Principle 6). In our own work, for example, we have begun to systematically investigate students' learning across different topics in the same field (e.g., bacteriophages vs. viral nucleic acids; Alexander, Pate, Kulikowich, Farrell, & Wright, 1989), across different fields of study (e.g., human immunology vs. physics; Alexander, Jetton, & Kulikowich, 1995), and under varying task conditions (e.g., linear vs. nonlinear text processing; Alexander, Kulikowich, & Jetton, 1994).

The outcomes of the studies just referenced and others (e.g., Cognition and Technology Group at Vanderbilt, 1990; Hidi & Baird, 1986, 1988; Schiefele, 1991; Wittrock, this volume) corroborate that a learner's knowledge can vary dramatically from topic to topic, from domain to domain, and from situation to situation (Murphy & Woods, 1996). For example, the child who demonstrates competence in one area (e.g., dinosaurs) or with one task (e.g., multiplication tables) may perform

like a novice in another area or with an alternate learning task (Alexander, in press-b). Likewise, the adult who is considered an expert in one domain (e.g., educational psychology) may best be described as a competent learner in some other domain (e.g., statistics) or even naive or acclimated in another (e.g., physics). This finding suggests that the knowledge base is not only complex but also fluid and dynamic in terms of its application. Thus, global and unconditional judgments of student capabilities and competencies on the basis of the assessment of the knowledge base may be unfounded and potentially detrimental to subsequent learning.

Strategic Processing or Executive Control

The ability to reflect on and regulate one's thoughts and behaviors is essential to learning and development.

Since the ground-breaking work of Flavell (1977) and Brown (1975), and with support of research in creative or critical thinking (e.g., Ennis, 1985, 1987, 1989; Nickerson, 1989), it is widely accepted that effective learners not only possess a body of organized and relevant knowledge, but they also have the ability, and at times the willingness, to reflect on and to oversee their own mental functioning and to assess their own performance (Principles 4 and 5). Studies have demonstrated that learning is enhanced when individuals have knowledge of and apply appropriate monitoring or executive strategies during the learning process (e.g., Elliott-Faust & Pressley, 1986; Garner & Alexander, 1989; Weinstein & Mayer, 1986). Whether these mental actions and self-regulations are specifically identified as general cognitive strategies (Pressley, Goodchild, Fleet, Zajchowski, & Evans, 1989), metacognitive strategies (Flavell, 1976; Garner, 1987), executive control or monitoring (Brown, 1975), executive components (Sternberg, 1980), or creative or critical thinking (Ennis, 1989), the findings are much the same. Those who reflect on their own thinking and learning performance and use that self-knowledge to alter their processing are more likely to show significant academic growth than those who do not (e.g., Garner, 1987).

Yet, the research in strategic processing or executive control has also undergone change over the past decades. Initially, there was lively debate in the literature over whether general or domain-specific strategies were more essential to learning (e.g., Glaser, 1984; Sternberg, 1985). The flurry of discussion around this debate has since quieted. It

is no longer a question of which is better. It is strongly held that both general cognitive and metacognitive strategies and domain-specific strategies are critical for learning (Alexander & Judy, 1988). The leading question has now become one of when, where, and for whom certain types of strategies are more effective for learning (Alexander, in press-b; Schoenfeld, 1988). In addition, there have been attempts to clarify the meaning of such fundamental terms as *strategies* (Alexander et al., 1991; Pressley et al., 1989). Even the notion of metacognition and creativity has undergone reexamination in recent years (e.g., Alexander, Parsons, & Nash, 1996; Gardner, 1993; Garner & Alexander, 1989; Paris & Winograd, 1990; Sternberg & Lubart, 1991).

Just as the research in knowledge has matured, so too have the investigations of strategic processing. For instance, during the height of research on metacognitive and cognitive strategies during the 1970s and 1980s, there was great hope that training in strategic processing or executive control would give rise to significant and long-term changes in students' learning. Yet, so many of those dreams were never realized. It became clearer and clearer to those engaged in this research and those who attempted to implement relevant findings in schools that strategic processing or executive control alone is necessary, but not sufficient, to ensure greater learning (Alexander & Judy, 1988; Garner, 1990). If training was limited in scope, not extended over time, or did not adequately address transfer to new tasks or situations; if the attributions and goals in the classroom did not support or reward strategic thinking; if students failed to see the relevance or value of the trained strategies; or if their self-judgments did not justify the effortfulness required of strategic behavior, then little if any real impact on learning could be expected from the teaching of cognitive or metacognitive strategies (e.g., Borkowski, Carr, Rellinger, & Pressley, 1990; Garner, 1990; McCombs, 1988; Palmer & Goetz, 1988; Paris & Winograd, 1990).

As with the dimensions discussed in the other principles, it is better understood that strategic processing or executive control must be coordinated with other factors, such as the learner's emotional state or the instructional climate, if optimal learning is to be achieved (Alexander, in press-a, in press-b). There is also greater understanding that learners' strategic processing should vary as their knowledge in a field grows and develops (Alexander & Judy, 1988; Hasselhorn & Körkkel, 1986; McCutchen, 1986). For example, McCutchen (1986) found that students' writing performance varied as their knowledge of content, discourse, and strategies changed. Likewise, Hasselhorn and Körkkel

(1986) investigated how instruction in a general knowledge activation strategy benefited children with differing amounts of content knowledge. Specifically, Hasselhorn and K rkel demonstrated that metacognitive training was most useful with novices, whereas a more traditional knowledge-activation study worked best with experts. On the basis of these results, the researchers concluded that there were “greater benefits of metacognitive instruction in selected settings” (p. 88).

Further, the reflection and strategic efforts must, almost by definition, vary as the demands of the task or the nature of the context changes (e.g., Lave, 1988, 1991; Pressley et al., 1992; Resnick, 1991). To assume that one can simply have students memorize and routinely execute a set of strategies is to misconceive the nature of strategic processing or executive control. Such rote applications of these procedures represents, in essence, a true oxymoron—nonstrategic strategic processing. It has become increasingly clear, as well, that strategic processing or executive control is not only a purposeful undertaking but an effortful one as well (Borkowski, Carr, & Pressley, 1987). This means that it is difficult to bring about changes in students’ strategic processing without addressing issues of motivation, interest, and self-regulation (Palmer & Goetz, 1988). This interplay among dimensions is a theme we return to later in this chapter.

Motivation and Affect

Motivational or affective factors, such as intrinsic motivation, attributions for learning, and personal goals, along with the motivational characteristics of learning tasks, play a significant role in the learning process.

Edison is credited with saying that “Genius is 1% inspiration and 99% perspiration.” Much the same can be said for learning in general, because learning is strongly influenced by the degree to which individuals are invested in the learning process. In other words, learning is altered by motivational and affective factors within the learner and in the learning environment. Indeed, many of the learner-centered principles from the APA task force (1993) relate to this dimension (i.e., Principles 2, 6, 7, 8, and 9). Whereas semantic issues about the construct of motivation persist (Pintrich, 1991), it is useful to view motivation as the general process by which the individuals’ needs and desires are activated and, thus, directs their thoughts and their behaviors (e.g., Pintrich, Marx, &

Boyle, 1993). Affect, as a related construct, can be understood as the feelings or emotions that individuals typically have and that, likewise, influence their thoughts and behaviors (e.g., Ames & Ames, 1985).

The research in motivation and affect has documented well over the decades that personal involvement, intrinsic motivation, and commitment contribute to greater learning (Principle 8; e.g., Ames & Ames, 1985, 1989; Corno & Rohrkemper, 1985; Dweck & Leggett, 1988; Gottfried, 1985, 1990). Those learners who have positive self-concepts (e.g., Wigfield & Karpachian, 1991; Wylie, 1974, 1979, 1989), believe themselves to be in control of their learning (e.g., deCharms, 1968; Deci & Ryan, 1985, 1991; Deci, Valleran, Pelletier, & Ryan, 1991), and have goals of understanding rather than performing (e.g., Dweck, 1986; Nicholls, 1984; Principle 2) are more apt to be those who achieve in schools. In addition, the students who are capable of setting appropriately high goals for themselves (e.g., Bandura, 1977, 1993; Schunk, 1991; Tolman, 1932) and those who find their academic work to be personally relevant (Principle 8) are more likely to succeed in schools (e.g., Lepper, 1988; Maehr & Braskamp, 1986; Meece, Blumenfeld, & Hoyle, 1988). The classroom environment in which students operate also plays a significant role in learning (Principle 6). When the teacher acknowledges students' personal goals and interests, or when learners perceive the academic climate as supportive and encouraging, they are more likely to perform well within the academic environment (e.g., Ames, 1992; Newman & Schwager, 1992; Pintrich et al., 1993).

Yet, motivational or affective influences are not restricted to general personality or emotional factors or to global orientations toward school or learning. These dimensions, as with the strategic factors we described earlier, have a domain-specific or task-specific character, as well as some general nature that is not clearly conveyed in the original principles. That is, although we can say that the research has supported the notion that the more motivated student performs better than the unmotivated student, we should also consider that no individual is equally motivated in all situations or contexts (e.g., Csikszentmihalyi, 1990; Schiefele & Csikszentmihalyi, 1994, 1995). To the contrary, learners are apt to show higher levels of motivation (e.g., interest or involvement) for some fields of study than others and for certain tasks but not for others (e.g., Alexander, Kulikowich, & Schulze, 1994a; Phillips & Zimmerman, 1990; Renninger, 1992).

These variations in motivation or affect may be attributed to a number of factors, including students' knowledge of the content (e.g., Al-

exander, Kulikowich, & Schulze, 1994b; Schiefele, Krapp, & Winteler, 1992), their perception of instructional priorities or teacher expectations (e.g., Alexander, Jetton, Kulikowich, & Woehler, 1994; Jetton & Alexander, 1997; Wigfield & Harold, 1992), or the perceived climate in the classroom (e.g., Ames, 1992; Schunk & Meece, 1992). In a recent study, for example, Schiefele and Csikszentmihalyi (1994) found that high school students' interest was significantly associated with the perceived quality of their experiences in their classes. In addition, the researchers reported that regression results suggested that performance in students' more difficult subjects (e.g., mathematics and physics) was more affected by ability, whereas performance in the easier to learn subjects (e.g., biology and social studies) was more influenced by motivational factors.

For the reasons just stated, there is an increasing trend in the literature on motivation and affect to investigate more specific dimensions of human learning and performance (e.g., Monroe, 1991). Thus, although the research on motivation and affect continues to flourish, there is a growing body of work in interest (e.g., Renninger, Hidi, & Krapp, 1992; Tobias, 1994), which, by definition, conveys the domain or task specificity that we have been addressing. That is, as with the Schiefele and Csikszentmihalyi (1994) study, we do not typically describe learners as globally interested, but rather as possessing interest in some subject, topic, task, or even text segment (e.g., Asher, Hymel, & Wigfield, 1978; Baldwin, Peleg-Bruckner, & McClintock, 1985; Hidi, Baird, & Hildyard, 1982; Pintrich et al., 1993; Schank, 1979).

One outcome of this changed focus in research on motivation and affect is the recognition that individuals' investment or involvement, even when positive, does not always result in a rise in learning. Indeed, under certain conditions, individuals' arousal or attention to information or factors that are not important or even tangential to academic goals can prove detrimental to the learning process (e.g., Garner, Alexander, Gillingham, Kulikowich, & Brown, 1991; Garner, Gillingham, & White, 1989; Wade, Schraw, Buxton, & Hayes, 1993). Garner et al. (1991) discovered, for example, that the presence of highly interesting, but relatively unimportant, information about a wager made by Stephen Hawking appeared to negatively impact students' ability to recall highly important but less interesting scientific content.

This recognition has important implications for educators, because it suggests that it is not sufficient for teachers to merely pique the attention or interest of their students without consideration of potential

ramifications. Rather, teachers' attempts to motivate must consider the personal goals and interests of the student (Dewey, 1913) and the relative value of the content (Wade et al., 1993), along with the desired outcomes of the instructional process (Alexander, in press-b). Further, the teacher must create an instructional climate in which the students have opportunities to engage their personal interests in meaningful ways and in a manner that acknowledges their personal control or regulation of their own learning (e.g., Pintrich & Schrauben, 1992; Zimmerman & Martinez-Pons, 1992). In sum, it would seem that the application of the literatures in motivation and affect to the reshaping and reformation of the educational system will need to consider not only learners' general needs, desires, self-perceptions, and emotional states or orientations but also their specific interests, goals, and desires.

Development and Individual Differences

Learning, although ultimately a unique adventure for all, progresses through various common stages of development influenced by both inherited and experiential/environmental factors.

Of the various dimensions of learning that we have analyzed, none has a longer or richer history than that of development. Indeed, the philosophical roots of traditional developmental theory reach well back to Plato, Rousseau, and Locke and are marked by attempts to understand the nature and growth of the human species. Although the philosophical foundations of individual differences may be relatively more recent (e.g., Luther, J. S. Mill), the resulting body of research is no less rich and impressive. The focus of this research is the variability that differentiates one person from another. As John Stuart Mill (1859/1947) so aptly stated, "The free development of individuality is one of the leading essentials of well-being" (p. 58).

In their articulation of learner-centered principles, the members of the APA task force (1993) saw fit to devote a specific guideline to development (Principle 10) and another to individual differences (Principle 12). This approach is quite understandable in light of the tendency among psychologists to think either in terms of global trends in human development (e.g., Case, 1985, 1993) or in terms of unique, individualistic patterns of change (e.g., Dillon & Schmeck, 1983; Reynolds & Willson, 1985). According to Scarr (1992), this separateness between development and individual differences is attributable, in large

measure, to the distinct philosophical traditions that underlie each, and to which we have alluded. Certainly, there is a rich and extensive literature that attests to progressive and predictable patterns in human development (e.g., Farrar & Goodman, 1992; Flavell, Mumme, Green, & Flavell, 1992; Zelazo & Reznick, 1991). Likewise, there is a plethora of research that speaks to variations in development at the level of the individual learner (e.g., Galton, 1908; Jensen, 1989; Plomin & Daniels, 1987; Plomin & DeFries, 1985).

Yet, there is no reason to assume that development must be conceptualized as either typical or individualistic in nature. Our goal here, therefore, is to reframe the original principles dealing with development and individual differences into one statement that conveys human growth as a process that entails not only commonalities across individuals but also unique variations within individuals. For instance, we can discuss the rather stable nature of IQ scores substantiated in populations over time, or we can describe the large variation in such scores that can be evidenced within individuals (e.g., Bayley, 1943; McCall, Applebaum, & Hogarty, 1973). According to Cole and Cole (1993), there are three questions that have marked the history of developmental studies. First, there are issues that concern the continuity or continuous development of the human species. Second, there are studies that focus on individual differences or the patterns that distinguish one person from all others, whereas, third, there are questions regarding the sources of development. For our purposes, we intentionally juxtapose discussions of typical population development against the theme of individual variability, thus, reducing our guiding foci to two: development (both group and individual) and sources of that development.

Of course, by choosing to combine these two original principles, we have undoubtedly made our task of analysis more demanding. However, we felt that this step was important so as to establish the perspective that learning is characterized by the continual interplay between the nomothetic and idiographic components of human development, that is, between the generalizable patterns that give human thoughts and actions more predictability and the individualistic constructions that are inevitable. Indeed, whether the focus is on cognitive (e.g., Case, 1985; Flavell, Miller, & Miller, 1993; Piaget, 1952; Vygotsky, 1934/1986), socioemotional (e.g., Barrett & Campos, 1987; Dabrowski, 1964; Dodge, 1980, 1986; Erickson, 1963; Hoffman, 1975; Skinner, 1953), or moral dimensions (e.g., Gilligan, 1977, 1982; Kohlberg, 1981; Wainryb & Turiel, 1993) of human development, educators must be knowledgeable of

the typical stages that mark the pathways through which individuals of shared characteristics (e.g., age, gender, or experiential background) appear to travel. Simultaneously, educators must not be blind to the fact that no two pathways of development are identical. Because every individual enters the world with a different genetic or biological history (e.g., Jenkins, 1979; Piaget, 1952; Plomin, 1986), particular psychological makeups, and idiosyncratic preferences (e.g., Csikszentmihalyi, 1990; Gardner, 1983), and because individuals develop within varied socio-cultural environments (e.g., Ogbu, 1974), their ultimate constructions are unique and nonreplicable (e.g., Nicholls, Chueng, Lauer, & Patashnick, 1989; Schmeck & Geisler-Brenstein, 1989).

This interplay between nomothetic and idiographic perspectives on human development was the basis for Scarr's 1991 Presidential Address to the Society for Research in Child Development. As Scarr noted, traditional developmental theories historically centered their attention on typical patterns of behavior and viewed deviations from those typical patterns as distractions (see, e.g., Chomsky, 1975; Luria, 1976). Reciprocally, other perspectives on human development, particularly some recent views, have seemingly discounted the commonalities that are basic to populations of learners and have chosen, instead, to focus solely on the unique constructions or personal realities that characterize individual thought and action (e.g., Tobin, 1991; von Glaserfeld, 1991). Still, there is no reason to assume that these dimensions of human development must remain separate, and there is good reason to expect them to become more integrated in emerging theories of development. Educational research and instructional practice can be well informed by both philosophical traditions and related literatures. This was exactly Scarr's (1992) intention when she proposed that

it is possible now to incorporate both typical development and individual variations on typical patterns in new theory that describe and explain human development as both human and uniquely so. I argue that most developmental theories today are unnecessarily limited in the observations they subsume and that they seldom address the issues of causality in development and individual variations with the same concepts. (p. 2)

If educators are to bring about the restructuring of schools that is the espoused goal of the APA task force, and of many other researchers and practitioners, then understanding and valuing of both the typical and unique dimensions of human growth and development must occur. These are indeed complementary, not contradictory or even separate,

dimensions of human growth. Thus, as with other dimensions we have reviewed here, it is not a matter of one perspective or another. Rather, optimal learning results when one accepts and appreciates the fluid and complex nature of human change. Perhaps our decision to combine the information on development and individual differences arises from occasions in our own teaching and research endeavors that have pointed out the potential ramifications of strict adherence to either of these two complementary perspectives. For instance, we have seen educators who make instructional and research decisions on the basis of notions about cognitive capabilities that resulted in the disregard of evidence to the contrary (e.g., Alexander, White, Haensly, & Crimmins-Jeanes, 1987; Alexander, Willson, White, & Fuqua, 1987; Gelman, 1969). In contrast, we have been witness to the instructional difficulties that emerge when novice teachers enter the classroom without an understanding of the general patterns that learners of particular attributes (e.g., age) are likely to exhibit (Alexander & Knight, 1993).

But what of the sources of development? Is growth and change in learners the consequence of inheritance or of experience? As the task force recognized in its learner-centered principles, human growth is influenced by both inherited and experiential factors. It remains debatable and empirically testable as to which of these powerful forces exerts the most influence on learners' growth. Depending on the specific aspects being examined, it may at times appear that changes in human thinking, feeling, or actions are more a reflection of one's inherent abilities, capabilities, conditions, or predispositions (e.g., Chomsky, 1986; Newport, 1991; Piatelli-Palamarini, 1980; Scarr, 1992). At other times, however, the research would seem to support the position that development is more than the outcome of one's life experiences and the context in which one learns and grows (e.g., Bronfenbrenner, 1986; Howes, Matheson, & Hamilton, 1994; Ogbu, 1974). By strategically acknowledging the potency of both heredity and environment, we and the task force avoid entering this debate and intentionally give both contributing powers their due.

Situation or Context

Learning is as much a socially shared undertaking as it is an individually constructed enterprise.

One of the most powerful observations that has emerged in the psychological literature in the past several years, and a premise that has

been woven throughout our discussion of the preceding dimensions, is the recognition that learning is continuously and markedly shaped by the social context in which it occurs (e.g., Alexander, in press-a; Cognition and Technology Group at Vanderbilt, 1990; Lave, 1988; Resnick, Levine, & Teasley, 1991; Rogoff, 1990). It is difficult to pick up a volume of research written in the past several years without encountering some reference to *socially shared cognition* (Resnick et al., 1991), *distributed intelligence* (Pea, 1988, 1989), *shared expertise* (Brown & Palincsar, 1989), *guided participation* (Rogoff, 1990), or some related terminology such as *situated action* (Greeno & Moore, 1993; Vera & Simon, 1993) or *anchored instruction* (Cognition and Technology Group at Vanderbilt, 1990). Principle 11 of the learner-centered psychological principles specifically acknowledges the social nature of learning and the value that shared thinking can have on learner performance. Indirectly, Principle 13, with its discussion of cognitive filters also attends to the external circumstances that are reflected in individual understandings and awareness.

Overall, what the literature underlying the work of the task force stresses foremost is the within-individual processes and conditions that are central to learning. This orientation is readily apparent in the task force's choice of title for its document—*Learner-Centered Psychological Principles: Guidelines for School Redesign and Reform*. Concurrently, the dimensions dealing with the knowledge base, strategic processing or executive control, motivation and affect, and development and individual differences discussed here are more concerned with the internal state of human learning than with social or contextual considerations. Yet, this internal state of the learner is in constant interface with external conditions and influences. With its inclusion of Principle 11, the task force encourages deeper examination of the impact of external forces on individual learning.

In many ways, the biological structure that separates the individual mind from the sociocultural context in which that mind functions can be thought of as a highly permeable fabric through which thoughts, feelings, and impressions move freely. Further, with each passage through this permeable fabric, those thoughts, feelings, and impressions are filtered and transformed, often without a conscious awareness on the part of the learner or those within the situation or context. Thus, educational change or reformation of the kind advocated by APA, the American Educational Research Association, as well as many other national associations and organizations (e.g., the National Council of Teachers of Mathematics and the National Research Council) cannot

be realized unless the power of the social or contextual factors involved in learning is also given credence (Yinger & Hendricks-Lee, 1993). Otherwise, the picture of learning remains largely incomplete.

As we see it, there are two general issues to be considered in this discussion of the socially shared nature of learning. The first relates to the potential effect of social interactions within the classroom, and the second pertains to the potential discrepancies in learning that can arise between in-school and out-of-school contexts. To understand the effect that social interactions and the learning contexts have on students' learning, one must first understand that schools are, after all, social institutions in which groups of individuals are brought together to share the educational experience. Students are in constant contact with peers and with other adults during the course of instruction. Rather than overlook this fact, various researchers have systematically investigated how these classroom interactions can be perceived or orchestrated to contribute to greater learning (e.g., Brown & Palincsar, 1989; Collins, Brown, & Newman, 1989; Johnson & Johnson, 1985; Palincsar & Brown, 1984; Slavin, 1978, 1987). As Resnick (1991) put it,

Recent theories of *situated cognition* are challenging views that the social and the cognitive can be studied independently, arguing that the social context in which cognitive activity takes place is an integral part of that activity, not just the surrounding context for it . . . every cognitive act must be viewed as a specific response to a specific set of circumstances. Only by understanding the circumstances and the participants' construal of the situation can a valid interpretation of the cognitive activity be made. (p. 4)

Although some socially interactive approaches, such as cooperative learning (Johnson & Johnson, 1985; Slavin, 1987), have been around for some time, a plethora of new orientations or instructional approaches have begun to appear in the literature that also advocate social interaction. Among these approaches are reciprocal teaching (Palincsar & Brown, 1984), cognitive apprenticeship (Collins et al., 1989), anchored instruction (Cognition and Technology Group at Vanderbilt, 1990), and Jigsaw (Aronson, 1978; Brown & Campione, 1990). Overall, the data on these various approaches have generally been favorable (e.g., Palincsar & Brown, 1984; Slavin, 1987). However, the empirical base for a number of these popular and theoretically appealing movements is still in an emergent phase. Educational researchers and practitioners, therefore, should continue their cooperative efforts to test the effectiveness of these and other approaches within the most appropriate laboratory for such research—the classroom.

Yet, learning, as we discussed earlier, does not occur only within the confines of the school or the classroom (Murphy & Woods, 1996). Learning takes place wherever the individual interacts with his or her world. Indeed, much of the influential works of Rogoff (e.g., Radziszewska & Rogoff, 1988; Rogoff & Gauvain, 1986) and Lave (1988) have documented the outcomes of socially shared learning among children and adults functioning in nonschool situations. Radziszewska and Rogoff (1988), for example, observed errand planning in 9-year-olds working with both same-age peers and adults. One of the most interesting results of this particular study was the researchers' determination that the 9-year-olds gained more skills in errand planning when they collaborated with a parent than with a same-age partner. It would appear that the knowledge and skills of those with whom a child collaborates or cooperates may be a significant component in the learning equation, both in and out of school.

Because the school, itself, is a unique and socially contrived context, there are instances when knowledge and skills acquired within that context do not transfer to other potentially relevant situations (e.g., Perret-Claremont, Perret, & Bell, 1980). Likewise, there is evidence that the knowledge and skills that individuals manifest in their daily lives fail to find their way into the school classroom (e.g., Lave, 1991; Rogoff & Lave, 1984). Lave (1991) argued that this lack of transfer is attributable to the absence of interconnectedness among communities of practice. It would appear that the artificial barrier between schooled and non-schooled contexts contributes to the inapplicability of valuable knowledge (e.g., Lave, 1988). Thus, the student who seemingly demonstrates competency in physics fails to apply that knowledge in out-of-school situations where it can be useful (e.g., diSessa, 1982). Reciprocally, the young man who can buy and sell baseball cards in expert fashion after school can actually appear rather mathematically incompetent when faced with school-like mathematics problems (e.g., Schoenfeld, 1985; Stewart, 1987).

One outcome of this growing body of research is the understanding that the context in which learning occurs is certainly nontrivial (Ber-eiter & Scardamalia, 1989; Cognition and Technology Group at Vanderbilt, 1990; Greeno, 1989; Greeno & Moore, 1993). It is also apparent that the role of the teacher as the facilitator and guide of these social exchanges is vital (Ellis & Rogoff, 1986; Radziszewska & Rogoff, 1988; Rogoff & Gauvain, 1986). In essence, when learners were able to benefit from the guidance of a knowledgeable adult (e.g., teacher) who pro-

moted their exploration and interchange, performance was found to be better. It has also been suggested, in line with Vygotsky's (1934/1986) concept of a zone of proximal development, that the guidance provided by the more knowledgeable and skilled adult be commensurate with the knowledge and skill of the learner. Thus, as the learner develops in an area or in relation to a given task, the direct or indirect assistance provided by the teacher should decrease proportionately (e.g., Alexander, *in press-b*; Brown & Palincsar, 1989). This lessening of external direction and support from teacher or adult should theoretically contribute to more independent functioning on the part of the student and, likewise, enhance the possibility of transfer of the acquired knowledge or skill both in class and out (e.g., Rogoff, 1990).

Issues for Further Consideration

As we stated at the outset of this chapter, the original set of 12 psychological principles (APA, 1993) we reviewed in 1994 have already undergone revision (APA, 1995). This revision was intended, in part, to address several important issues that we and others believed were only indirectly addressed in the original learner-centered principles, or not specifically addressed at all. Those issues, some of which we alluded to previously, include an attention to the multidimensional, interactive character of learning and a recognition of the domain-specific or context-specific nature of learning. Further, we wanted to consider the issue of assessment, not in terms of its role in measuring student outcomes or progress, but in terms of its contributions to planning more effective curricula and to building optimal learning environments for students. Finally, we caution educational researchers and practitioners against falling victim to premature and overly optimistic support of emerging and largely untested ideas or approaches, regardless of their political power or their intuitive appeal.

Interactive Dimensions of Learning

In its preface to the original learner-centered principles, the APA task force stated that the principles they championed "must be understood

as an organized set of principles and not be treated in isolation” (APA, 1993, p. 6). We concur with that statement, but we wish to extend it somewhat. Although the task force may say to educators that the learner-centered principles should not be dealt with in isolation, the specific format and wording of the report may unintentionally reinforce that perception. That is, among the original principles, there are several that speak to metacognitive and cognitive factors, affect, personal and social factors, and individual differences, and another that overviews development. Of course, the pragmatic and theoretical rationales for quartering off learning into foundational areas are certainly justifiable—particularly in light of the massive literatures on which each is grounded. For those awarded the honor of substantiating these principles from the research base, that rationale is evident indeed. However, even as we consciously dissect learning into its essential dimensions, we must strongly acclaim the serious limitations of this organizational strategy.

Specifically, if meaningful change is to occur within the educational system, then it must be better understood by educators and educational researchers that these dimensions of learning are not really separable entities in “real-world learning situations” (APA, 1993, p. 6). Although, as researchers, we can theoretically and empirically extract cognition from affect, knowledge from strategic processing, or sociocultural background from development, these dimensions, as with the others addressed in the principles, remain inextricably intertwined in the real world. For example, the knowledge base is both an aspect of development (e.g., Chi, 1985; Chi et al., 1981) and a part of motivation (e.g., Renninger, 1992; Schiefele, 1991), just as development is integrated with individual differences and sociocultural background (e.g., Ogbu, 1974; Scarr, 1992) and strategic processing is correlated with one’s motivational or affective state (e.g., Garner, Brown, Sanders, & Menke, 1992; Hidi & Anderson, 1992; Weinert & Kluwe, 1987).

By approaching learning in a systemic fashion, attending simultaneously to the interplay of its various dimensions, educators perhaps have a greater chance to make significant and long-term changes in students’ learning and in the institutions created especially to promote such learning (Salomon, 1991). In his analysis of human behavior and the experiences from which it arises, Dewey (1930) expressed this precept: “No act can be understood apart from the series to which it belongs” (p. 412). We could not agree more.

The Domain-Specific Nature of Learning

As we have noted at various points in our analysis of the research literature, the subject matter with which a student deals, as well as the learning tasks and the contexts in which that individual operates, must be given its rightful due, not only in the learner-centered principles but also in the educational practice that these principles seek to impact. This domain and context specificity is addressed to some extent in the task force's presentation of the implications of the learner-centered principles for school redesign and reform. However, this orientation is less apparent in the principles themselves, which are likely to be discussed and highlighted separately from any implications the task force has chosen to advance. Therefore, we would prefer to see this issue of domain and context specificity more directly considered in each of the relevant principles.

For instance, individuals' knowledge base and their motivation to achieve or to learn have a significant impact in the process of schooling. However, we cannot overlook the fact that knowledge, motivation, and the other dimensions that we have described here are highly complex and varied phenomena. Whether students are asked to function in a subject matter area for which they feel competent or for which they have an abiding interest versus one for which they possess only fragmented knowledge, or little if any interest, is critical (Alexander, *in press-b*). Teachers should not expect students to be consistently and invariantly knowledgeable or motivated in classrooms. In addition, teachers should have a realization that the motivational character and the meaningfulness of tasks and learning environments are not solely inherent in the tasks or environments, themselves. Rather, these positive attributes are as much a consequence of the knowledge and interest that the learner brings to that task or that environment. Further, as Spiro argued (Spiro, Feltovich, Jacobson, & Coulson, 1992; Spiro & Jehng, 1990), domains of knowledge are different, and those differences should be attended to as educators attempt to enhance student understanding in classrooms. For example, learning physics is unique from learning literature or social studies (Alexander & Kulikowich, 1994; diSessa, 1989; Larkin, 1985; Larkin et al., 1980). By identifying the specific attributes and concomitant challenges that each domain may present to students, and students' knowledge and interests relative to each domain, teachers will be better equipped to organize the instructional tasks and the learning environment more effectively.

After our previous discussion about the interactive and multidimensional nature of learning, we now turn to the domain-specific nature of learning.

mensional nature of learning, we think it necessary to explain further our call for more consideration of the domain-specific nature of learning. Specifically, the acknowledgment of domain-specific influences in the learning process has importance for two reasons. First, it is our contention that an individual's progression from a naive or acclimated learner to one who is competent and, potentially, even expert seems reliant in large measure on his or her ability to function with a "principled" understanding in a given domain (Alexander, *in press-b*; Gelman & Greeno, 1989).

Principles can be thought of simply as the guiding concepts or understandings that give a domain its particular configuration or structure. In mathematics, for example, one such principle deals with the "conservation of number," whereas in reading, one guiding principle relates to the "meaningfulness of text." Until learners grasp these fundamental understandings, their acquisition of relevant information in the domain is apt to be fragmented and incoherent. Therefore, domain-specific principles become organizing concepts for students' knowledge base. According to Gelman and Greeno (1989), the presence of domain-specific principles serves another role in the learning process. These principles provide learners with a mechanism for ascertaining the relevance of new information. In addition, Gelman and Greeno hold that principles offer clues to the interpretation of ambiguous information within the environment—information that has the potential to either foster or frustrate learning.

Our second purpose for stressing the domain-specific characteristics in learning is related to the first. Specifically, as educators pursue the laudable goal of integrating bodies of knowledge within the learning environment (as in the case of the dimensions of learning we analyze here), they should appreciate that this integration is aided by students' ability to build meaningful connections across those bodies of knowledge (e.g., Holyoak, 1991). Yet, to build connections across two or more areas of knowledge, learners must possess some clear conceptualization of each contributing factor. For example, to see the relationships that exist between early American literature and the historical events of that time, students must have or be given both relevant literary and historical information around which these relationships can be forged. In essence, the ability to see relationships across seemingly diverse areas of learning requires awareness of what makes each area unique in and of itself. Otherwise, efforts to create a beneficial conflu-

ence of bodies of knowledge can, regrettably, result in little more than a hodgepodge of unusable and disjointed content.

The Instructional Value of Assessment

Although multiple chapters of this volume (e.g., Paris, this volume, chap. 8) are dedicated to the various aspects of assessment, learner assessment is not treated as a statement of principle in the original publication of the learner-centered psychological principles. We found this particularly interesting in light of the fact that more space was devoted to the discussion of assessment than to any other issue. Perhaps it was the judgment of the task force that assessment did not qualify as a within-individual psychological factor appropriate for consideration as a principle of teaching and learning. However, the attention to “relevant, authentic learning tasks” (APA, 1993, p. 7) would seem to open the door to discussions of assessment, especially as it pertains to creating effective learning tasks and optimal learning environments. Nonetheless, in light of the weight given to assessment in American classrooms and to the political and philosophical debates that this topic engenders, some specific statement of principle about assessment seems warranted. If we were to incorporate a principle relative to assessment, it might read as follows:

The development of relevant learning tasks and the orchestration of optimal learning environments is predicated on access to accurate and useful information about such dimensions of learning as students’ knowledge base (e.g., sociocultural background and beliefs), their strategic processing, as well as motivational and affective factors (e.g., interests and preferences).

We are pleased to note that the revised version of the document (APA, 1995) now contains a specific principle addressing the topic of assessment (Principle 14). Moreover, this principle incorporates and extends our proffered statement.

Our goal from a psychological perspective in this discussion is to alert the educational community to the fundamental point that assessment does not simply equate to the evaluation of student achievement nor to standardized measures. Instead, assessment should be viewed simply as the gathering of pertinent information in a systematic and useful fashion. Sometimes that information concerns the outcomes of the instructional process or includes formal tests or standardized measures of achievement, aptitude, personality, or other dimensions of learning. However, this is not the only intent or nature of assessment.

When teachers gather information on student interests, analyze responses to oral or written questions, gauge strategic processing, or monitor student reactions to particular learning conditions or tasks, they are engaged in assessment. What separates relevant from nonrelevant assessment then becomes a matter of the appropriate interpretation and application of the assembled information. Thus, *assessment*, defined in this admittedly liberal manner, should be seen as integral to effective teaching and learning and should not be held as something apart from them (e.g., Alexander, Kulikowich, & Jetton, 1994; Kulikowich & Alexander, 1994; Weinstein & Meyer, 1991; Wittrock, 1991).

We realize that some of these same concerns and perspectives were contained within the 1993 task force presentation of the implications and applications of the learner-centered principles. In addition, to our knowledge this topic has received much deserved attention during the revision of these principles, as evidenced by the addition of the specific assessment principle (14). Our issue is that the distinction between assessment, as broadly conceived, and evaluation, as narrowly portrayed in certain educational or political circles, needs to be made more emphatically.

Cautionary Note

Perhaps in a far too idealistic fashion, we would hope that research of whatever paradigm or perspective would serve as a voice for reasoned and informed practice, rather than as an inciter of untested or potentially risky actions. It is critical that the educational research community not become part of the very problems or conditions it purports to study (Alexander, Murphy, & Woods, 1996, 1997). If practitioners are to take researchers and their findings seriously, then it is mandatory that the educational research community establish and maintain credibility with the community of practitioners that must bring our ideas to light in the instructional setting. If the judgments and stances of the research community are perceived to shift too easily or too willingly with the winds of political pressure or in the direction of external funding, then it is likely that this credibility will suffer. Although it is not necessary that they always lead the way in instructional practice, educational researchers should at least share their knowledge with that leadership.

Therefore, as partners in school reform and redesign, educational researchers must be careful not to fuel the fires of debate or ignite passions when such emotionalism is unjustified or unsupported. This is

not to say that educational researchers should not be proactive and vocal when issues are clearly warranted, as in the case of the extended discussion of assessment found in the task force report. Where we wish to take issue is with the choice of wording in several of the principles—words that project a particular bias or orientation that is, in our opinion, unsubstantiated or unclarified from the perspective of the research. We focus on just one principle to illustrate our argument.

In the original principle dealing with learning tasks, the task force stated that “curiosity, creativity, and higher order thinking are stimulated by relevant, authentic learning tasks of optimal difficulty and novelty for each student” (APA, 1993, p. 7). The gist of this statement would seem to be that better instructional tasks produce better learning. However, the presence of such terms as *authentic* give this principle a character that becomes more difficult to substantiate. Is the literature clear on what constitutes an “authentic” task? As the term is sometimes used in the literature, is there an evident causal relationship between authenticity and student curiosity, creativity, and higher level thinking? Are all tasks that fall outside the rubric “authentic” assumed to be nonauthentic or fraudulent? Are there, as the task force suggests in its introductory remarks, decades, if not “more than a century” (APA, 1993, p. 5) of research to uphold this claim? As our position in this discussion suggests, we contend otherwise.

Indeed, as we stated at the outset of this chapter, we recognize that any report of a committee, such as the task force document, represents a compromise among many individuals from many perspectives. It is likely, therefore, that the inclusion of such emotionally laden terms as *authentic* is an outcome of this compromise. We feel that the goal of educational reform and redesign would have been better served by offering a more bottom-line analysis of learning that indeed has a long history of rather consistent outcomes to substantiate it. (The current version of the learner-centered psychological principles prepared by the APA Board of Educational Affairs, 1995, does not include this reference to authentic tasks.)

Conclusion

Again, we support the APA Presidential Task Force on Psychology in Education (APA, 1993), as well as the APA Board of Educational Affairs (APA, 1995), in their attempts to interject psychological principles into

current efforts to reform and redesign schools. Educational practice certainly can benefit from the valuable and essential knowledge that has been contributed by generations of educational researchers. As our analysis demonstrates, there are indeed massive bodies of research that uphold the role that the knowledge base, strategic processing or executive control, motivation and affect, development and individual differences, and situation or context play in student learning. Sadly, too little of that research has made its way into the public and political mainstream and, likewise, into American classrooms. Perhaps the efforts of the task force will help alter this regrettable situation. Yet, some equally powerful statements authored by the research community in the past have fallen short of their promise to bring about relevant change in instructional practice (e.g., Anderson, Hiebert, Scott, & Wilkinson, 1985). This observation leaves us cautiously optimistic about the prospects for the future with respect to the goals of the task force. Of course, the future is not yet written, and the concerted efforts of the educational research community, in partnership with others who are invested in the learners of today and tomorrow, can possibly make the hopes expressed in the learner-centered principles a reality. As individuals who share these hopes, we anxiously await the outcome.

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